

Undetectable but Unremovable: Capacity versus Topology in Lossy Networks

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DRAFT v0.1 — 2026-06-11. Paper 3 of the program. Genre: experimental mathematics / position paper; every claim compiles, cites, or wears its CONJECTURE/EVIDENCE tag (protocol of 00-master-plan.md). Written draft-first per Carlos's 2026-06-11 steer; paper3-outline.md is the compression, paper3-pseudohaskell.md the type audit. Numbers from mythos-notebook Entries 3, 5, 10, 14, 15. [REF:] marks the citation pass. IP-freeze: local artifact; nothing outward. Currency pass 2026-06-13: the magnetic-translation pair + its m-uniform commutator bound are now machine-checked (Entries 24–25); the dilation gate locates the open part as the index's rank-uniform stability (existence reduces to the classical winding); local-defect tracking extended to $m = 32$; partial F1 stealth instance in hand. Currency pass 2026-07-01 (v0.2): Stage B now has a VALID machine-checked reduction — plateau_of_rigidity proves the conjecture from the single estimate FluxPairRigidity (dimension-free rigidity of the flux pair against every commuting contraction pair); the pair-level adversarial kill test (e4) did not fire (no commuting pair found below the trivial zero-pair distance of exactly 1, any $m = 3 \dots 10$ or rank-padding; the trivial pair caps the pair-level constant at 1 — corrected same day after first write-up misread optimizer output above 1 as a floor). §3/§7 status notes updated. NOTE also the companion three-gaps result (speculativepaper5 §4a): the edge-level constant is codimension-dependent — $\sqrt{2}$ at codim 1, exactly 1 at codim ≥ 2 — so this paper's " $c(1) \approx 1.5$ " empirical figure is the codim-1/optimizer-bound reading; the proved codim ≥ 2 value is 1, which only strengthens $c(k) \leq \delta^$ (1 is still $\leq$ any realistic threshold). IP-freeze footer corrected (lifted 2026-06-13). v0.3 (2026-07-03): the §4a codimension constants folded through the body, which had still quoted the codim-1 fit-*

ted figure alone — abstract, §3 Stage B box, §5 worked example (now led by the proved floor of exactly 1, $35 \times \delta^*$, with the fitted 1.655 demoted to a missed-basin upper bound), §5 gap sentence, §8(3) (k -independence at $\text{codim} \geq 2 \implies$ the stealth window widens with charge). v0.3b (same day, later): Zhang/Joshi $\square$ RESOLVED — provenance was on disk all along (ct4-skeleton.md 2026-06-11: “Source: Joshi arXiv:2305.10398v12 §10 ... reading Zhang 2001 / Amorós et al. 2000 through Joshi’s presentation”), independently content-verified online (zhang-joshi-candidates.md: Zhang §3 matches §7’s dictionary line-for-line; Joshi’s §10 is literally titled “The proof of geometric Szpiro inequality revisited”). Bonus catch: the §7 bracket’s “Amorós-Burger-Corlette-Kotschick-Toledo” was a mis-resolution of the notes’ “Amorós et al. 2000” = Amorós-Bogomolov-Katzarkov-Pantev (JDG 54, 2000) — corrected in bracket + references. F2 re-rendered to the proved floor (solid $c = 1$; fitted 1.655 dashed as missed-basin upper bound). v0.3c (same day, evening): F1 battery RUN — paper3_f1_stealth.py/.png: planted unit charge, $m = 5 \dots 32$; at every m the certificate reads AUC 1.000 while naive and patch local monitors sit at chance (0.45–0.53); local signature crosses $\delta^* = 0.028$ exactly at $m^* \approx 15$ (at $m = 16$: 0.0245, the worked example’s $2\pi/256$); floor 1 = $35 \times \delta^*$ pinned. Synthetic/planted, labeled as such. No open figure debts remain for this paper.

Abstract

A network of lossy translations — partial isometries that destroy information as they carry it — can host a topological charge: a winding in the surviving subspace whose local signature shrinks with system size while the global cost of removing it does not. A companion paper established this plateau empirically and conjectured its mechanism; a separate validation exercise established, on real composed pipelines, the operational detection threshold of a noisy, quantized estimation chain. This paper puts the two results in one inequality. The local evidence of a charge k on an $m \times m$ torus is a per-plaquette defect of size $\approx 2\pi k/m^2$, which falls below the operational detection threshold $\delta^* = \max(1.405 \cdot \sigma\sqrt{\square}, 0.6 \cdot B/\sqrt{n})$ at a computable crossover size m^* ; beyond m^* , no local monitor at that instrumentation can see the charge. The cost of removing the charge, by contrast, is bounded below by a constant $c(k)$ independent of m , whose height is now codimension-resolved (companion three-gaps note, §4a): $\sqrt{2}$ where the surviving subspace has codimension one (measured, converging to $\sqrt{2}$ from below), exactly 1 at codimension ≥ 2 (proved for $r = 2$) — either way dozens of times above realistic thresholds. Past the crossover the network carries a *stealth charge*: undetectable by every local test, unremovable by any local edit. We give the precise statements with their evidence status — the existence floor is a theorem at every fixed size (Stage A; elementary proof, Lean formalization queued); the m -uniform constant is the companion paper’s conjecture (Stage B) — derive the crossover, work a numerical example, and map what falsifies what. The claim’s proof shape — one topological quantity evaluated two ways, locally as a sum of vanishing defects and globally as a protected obstruction — is the shape of the geometric Szpiro inequality, and we close with that dictionary and its one honest discrepancy.

1. Three silos, one question

Three literatures hold the pieces of this paper, and do not currently speak.

Stability of almost-representations (mathematics): when is an almost-satisfying map close to an exactly satisfying one? The modern landscape is sharp on the (norm $\times$ defect-functional) grid — Frobenius-pointwise stability from cohomology vanishing

[REF: DCGLT], operator-pointwise *instability* through index obstructions [REF: BLSW; Dadarlat; Voiculescu; Exel-Loring], uniform stability in all submultiplicative norms [REF: GLMR]. Everything assumes invertible targets.

Topological protection (physics): an integer invariant — a Chern number, a Bott index — cannot change under perturbations small in operator norm, so locally almost-trivial data can be globally far from trivial [REF: Voiculescu; Exel-Loring; Loring-Hastings]. The invariants live on unitaries or invertibles.

Detection theory (engineering): an estimation chain of depth $\square$ with per-stage noise σ , quantizer bandwidth B , and n samples cannot resolve effects below an operational threshold; classical ingredients (d' index, non-regenerative cascade, resolution floor) [REF: detection-theory texts; GUM] compose into the two-regime law validated — by pre-registered falsification of its product-form alternative — in our pipeline study: $\delta^* \approx \max(1.405 \cdot \sigma \sqrt{\square}, 0.6 \cdot B/\sqrt{n})$. **[EVIDENCE: E3]**

The question none of the three asks: **when is a topological charge in a network of lossy links observable at all** — and what happens in the gap between “too small to see” and “too protected to remove”? The companion paper (Paper 2) moved the stability question to lossy targets, where translations cannot be undone, and found empirically that the index obstruction survives rank loss. This paper assumes that result at its evidence level, adds the detection side, and states what the junction implies. We call the underlying quantity *semigroup holonomy*: holonomy carried by composable but non-invertible maps.

2. The objects (imported, with their tags)

We work with the companion paper’s torus systems; full definitions there and in the Lean formalization (LossyCocycles/Defs.lean — the definitions below are machine-checked objects, not prose).

- **Lossy cocycle.** Each oriented edge of the $m \times m$ torus carries $T_e \in \mathbb{R}^{d \times d}$ of rank $\leq r$. *Flat* (exactly consistent) families are exactly $T_e = V_j V_i^\top$ for orthonormal frames V ; every cycle composite is then a projection. [THEOREM, Lemma 1 ($\Leftarrow$) machine-checked; ($\Rightarrow$) prose appendix of Paper 2]
- **The flux family.** A Landau-gauge connection with total charge k carried in the $SO(2)$ core of the surviving r -subspace; per-plaquette angle $2\pi k/m^2$. Its local defect is $\delta_O \approx 2\pi k/m^2 \rightarrow 0$; this is elementary and measured (δ_O tracked $2\pi/m^2$ exactly over $m = 8 \rightarrow 16$, extended to $m = 32$ in `e_radar_discriminator.py`: $0.393 \rightarrow 0.006$). [THEOREM-grade arithmetic + EVIDENCE: E2-v2, m-scaling]
- **Distance to flat.** `distOp` = inf over flat families of the worst-edge operator distance. Fits give upper bounds; four guards (noise-floor control, invertible-anchor reproduction, planted-class non-absorbability, Frobenius-decay of the same solutions) plus an oracle-initialized kill test protect the lower-bound-shaped readings. [EVIDENCE: Entries 3, 10]

The plateau. For $k = 1$ and $r \in \{2, 3, 4\}$: δ_O falls $7\times$ as m : $3 \rightarrow 8$ while ε_O stays pinned at 1.29–1.66, and stays flat to $m = 16$ ($1.659 \rightarrow 1.682 \rightarrow 1.655$ at $r = 2$; ratio 1.00 across a $4\times$ sweep; pre-registered call UNIFORM); the same solutions are Frobenius-close. The obstruction is irreducibly 2-dimensional: a per-column argument is capped at γ/m by contraction telescoping, which at $m = 16$ would permit $\varepsilon_O \approx 0.4$ against a measured

1.655. [EVIDENCE: E2-v2 + Entry 14; the 2D-coupling necessity is a measurement, not a heuristic]

3. The two evaluations, and the Stage A / Stage B architecture

The proof-shape of everything below is: **evaluate one topological quantity two ways.**

Local evaluation. The total winding $2\pi k$ is the sum of m^2 plaquette angles, each $2\pi k/m^2$. Locally the charge is everywhere and almost nothing: any single measurement sees at most a $2\pi k/m^2$ -sized deviation from flatness.

Global evaluation. The same winding, read as an obstruction class, forces a lower bound on the distance to flat. This splits into the two stages adopted in this program's Response 27 (after external structural feedback):

Stage A (existence, fixed m) — THEOREM, machine-checked (`stageA_skew_bound`, `axiom-clean`). For $m > 2|k|$, every flat family is at worst-edge operator distance $\geq |\sin(2\pi k/m)|/m \approx 2\pi|k|/m^2$ from the charge- k flux family. *Mechanism:* flat column composites are projections, whose core skew is exactly 0; the flux column composite has core skew $2 \cdot \sin(2\pi kx/m)$; skew is entry-dominated by the operator norm and spreads at most m -fold along a column. No winding apparatus needed.

Stage B (uniformity) — CONJECTURE (the companion paper's). $\text{distOp} \geq c(k) > 0$ independent of m and $r \geq 2$; empirically $c(1) \approx 1.5$ (plateau 1.29–1.66; kill-test floor 1.49) — a **codimension-1 reading:** the three-gaps note (§4a) resolves the height as $\sqrt{2}$ at codim 1 (measured, monotone up to $\sqrt{2}$) and exactly 1 at codim ≥ 2 (proved for $r = 2$; the binding competitor there is orthogonal, and fitted readings above 1 are missed-basin artifacts), so the m - and r -uniform floor the conjecture needs — and the number this paper's inequality uses — is 1. Formal-route correction (2026-06-12): the natural first guess — an m -uniform winding-rigidity lemma over the column-path winding — is **provably wrong:** the shear family (flux verticals, trivial horizontals) winds k within $2\pi|k|/m$ of flat with full gap (`Shear.lean`, machine-checked). The column-path winding reads no horizontal data and is a 1D shadow, not an obstruction. The viable route is the Bott index of the magnetic-translation pair, whose commutator is the *sup* of plaquette defects (edge-local, m -free) — **the pair and that bound are now built and machine-checked** (`MagPair.lean`, Entries 24–25: $[U, V] = 0$ exactly on flat families; $\|[U, V]\| \leq 2|\sin(\pi k/m^2)|$ on the charge- k flux family, m -uniform, `axiom-clean`). What stays open is the **index of that almost-commuting pair and its Exel-Loring stability** — and a dilation analysis (2026-06-13) places that gap precisely: the index's existence is the classical 2-D winding (r -independent, `totalFlux_flux`; dilation-trivial), so the open, non-classical content is exactly the m -and- r -uniform stability for partial isometries — Stage B itself.

The two stages are different mechanisms, and Stage A's constant says so: it decays like $1/m^2$, exactly the size of the local evidence. **One-dimensional analysis certifies precisely what local observation can see, and no more.** The plateau — the global floor — is invisible to both. Stage A is therefore *not* evidence for Stage B and is not offered as such; it is the boundary marker of the local mechanism. (Status note, updated 2026-07-01: `stageA_skew_bound` is landed and `axiom-clean`, and Stage B now has a

valid, machine-checked reduction — the first candidate reduction was falsified by the shear family before anything was built on it; the corrected route then landed in stages: the magnetic-translation pair, its exact flat commutativity, and its m -uniform commutator bound (Entries 24–25), and finally `plateau_of_rigidity` (`PlateauConjecture.lean`, 2026-07-01), which proves Stage B from ONE scoped estimate, `FluxPairRigidity`: a $\delta(k) > 0$ independent of m , d , r separating the flux pair from every commuting contraction pair. That estimate — the transported dimension-free Exel-Loring/Voiculescu rigidity, uniform across rank deficiency — carries the project’s single sorry, and is elementarily capped at $\delta(k) \leq 1$ (the zero pair commutes and sits at distance exactly 1). Its direct pair-level kill test (e4: adversarial optimization over commuting contraction pairs with exact-commuting certification) did not fire: no competitor below the trivial distance 1 was found at any $m = 3 \dots 10$ or rank-padding — though the search never beat the trivial basin either, so this is a non-firing kill condition, not a measured floor.)

4. The detection side

The validation study E3 measured what a composed estimation chain can operationally distinguish, with a pre-registered gate that its product-form alternative failed:

$\delta^* \approx \max(1.405 \cdot \sigma \sqrt{\square}, 0.6 \cdot B / \sqrt{n})$ **[EVIDENCE: E3, two-regime law; regenerative bonus at coarse quantization noted there]**

σ = per-stage noise, $\square$ = chain depth, B = quantizer bandwidth, n = samples. Two classical constraints, never simultaneously operational; the deliverable was the *max*, replacing the provisional product form.

Extrapolation risk, named: E3 was validated on scalar chains. Applying δ^* to the estimation of a plaquette holonomy angle treats the monitor’s scalar readout (an angle or a residual) as the chain’s carried quantity. This is the weakest link in this paper and is tagged accordingly; the matrix-valued version of E3 is an open experiment (§6, F2).

5. The inequality, the crossover, and the stealth regime

A local monitor at a plaquette estimates a deviation of size $\approx 2\pi k/m^2$ (in operator units: $2|\sin(\pi k/m^2)|$, the vortex-core form). It sees the charge iff that exceeds δ^* .

Crossover. $2\pi k/m^2 \geq \delta^* \iff m \leq m^* := \sqrt{(2\pi k / \delta^*)}$.

The claim (the paper, in one box). On an $m \times m$ torus of lossy links with instrumentation $(\sigma, \square, B, n)$ and charge k : - **$m \leq m^*$** : the charge is locally detectable, and unremovable at cost $< c(k)$. - **$m > m^*$** : *stealth regime* — every local signature sits below δ^* , so no per-plaquette test at this instrumentation detects the charge [EVIDENCE-grade, inherits E3’s tag]; yet flattening the network still costs $\geq c(k)$ at some edge [CONJECTURE, Stage B; $\geq |\sin(2\pi k/m)|/m$ THEOREM, Stage A], and $c(k) \square \delta^*$ in every measured cell.

Undetectable is not removable. The boundary m^* is computable from instrument constants alone.

Worked example ($k = 1$). Take $\sigma = 0.01$, $\square = 4$ (noise term $1.405 \cdot 0.01 \cdot 2 = 0.028$); $B = 0.1$, $n = 100$ (resolution term $0.6 \cdot 0.1/10 = 0.006$). Then $\delta^* = 0.028$ and $m^* = \sqrt{(2\pi/0.028)} \approx 15$. On a 16×16 torus — exactly the size where the plateau was measured flat at

1.655 — the unit charge’s local signature is $2\pi/256 \approx 0.0245 < \delta^*$: invisible to this instrumentation. Its removal cost: the retraction-proof number is the proved $\text{codim} \geq 2$ value of exactly 1 (three-gaps §4a; $r = 2$, this cell’s rank), already **thirty-five times the detection threshold** — the fitted 1.655 is a missed-basin upper-bound reading above it. A maintainer who trusts local diagnostics would certify this network consistent and then find that no small recalibration of any link — or of all links — flattens it.

What the stealth regime is not. It is not an artifact of weak instruments: improving σ , B , n raises m^* but never closes the regime ($\delta^* > 0$ for any physical chain, and the $2\pi k/m^2 \rightarrow 0$ decay always wins). It is not a finite-size effect: Stage B says the unremovability is size-uniform; only its *detectability* is size-bound. And it is not mysterious: both sides are classical (detection theory; index rigidity) — the content is that they meet with a gap of one-and-a-half to two orders of magnitude ($35\times$ at the proved floor in the worked cell), in a sector (lossy targets) whose new content is **not** the index itself (a dilation reduction, 2026-06-13, shows it is the classical 2-D winding with index-0 padding, r -independent — `totalFlux_flux`) but the **rank-uniform stability** of that index for non-invertible maps (Stage B).

6. Falsification map

if this fails	then
Stage B (a flat family with $\varepsilon_O \rightarrow 0$ exists — or, sharper post-reduction, commuting contraction pairs approaching the flux pair)	$c(k)$ dies; the claim degrades gracefully to Stage A’s finite- m floor: the stealth window becomes $[m^*, M(\varepsilon)]$ — bounded, not infinite. The paper’s headline dies; its finite-size form survives as theorem. (Directly attacked 2026-07-01 at the pair level, e4: did NOT fire — no commuting pair found below the trivial zero-pair distance of exactly 1, at any $m = 3\dots 10$ or rank-padding. Note the trivial pair caps the pair-level constant at 1.)
E3’s law off scalar chains (F2 experiment: matrix-valued monitor)	δ^* needs re-measurement; m^* moves; the <i>structure</i> (some $\delta^* > 0$ exists for any physical monitor) survives on information-theoretic grounds.

if this fails	then
$r = 1$ boundary (charge with no $SO(2)$ core shows a plateau)	the index reading of the plateau is wrong; CT-4 and §3 both die. (Cell RUN 2026-07-03, paper2_r1_boundary.py: the trigger as worded did NOT fire — no charge exists to plant at $r = 1$. What was found instead: the curvature-free $Z/2$-wound family plateaus at every m — fitted 2.01, explicit absorption family $\rightarrow 1$ from above — with an elementary alignment-rigidity lower-bound sketch. The index reading stands; the prediction that needed correcting was §8(5)’s “no stealth at $r = 1$”: the stealth phenomenon exists there too, carried by $Z/2$, at height exactly 1 — PROVED 2026-07-04** (../paper2-r1-theorem-draft.md: closed-form alignment costs; both bounds by hand; any graph, any m , $d \geq 2$; Frobenius $\sqrt{(2/m)}$ exact), the same orthogonality barrier as the $\text{codim} \geq 2$ constant, by the same technique.)** FIRED 2026-06-12, machine-checked (Shear.lean): the column-winding proof program is dead; the conjecture needs the flux-aware Bott-pair index — graceful degradation exactly as this row promised. Evidence (E2-v2 on flux) untouched.
kill criterion of CT-4 (a winding-changing, gap-preserving path)	

Planned experiments, value order: **F2** matrix-valued δ^* (the named weak link); **F1** the stealth-regime demonstration end-to-end (plant charge at $m > m^*$, run the *actual* E1-style local certificate battery, show null detection while the fit cost stays $\approx c(k)$) — this is one script away from existing tooling; a synthetic instance is already in hand (DARPA-maybe/e_radar_discriminator.py: at $m = 32$ the local defect 0.006 sits $\approx 16\times$ below a $\sigma = 0.10$ noise floor while the global flux stays pinned at 2π , certificate AUC 1.000 vs a chance-level 1-D monitor — the stealth signature, on a radar phase network; the E1-style battery remains the full F1); **F3** k-sweep of the crossover ($m^* \propto \sqrt{k}$ prediction).

7. The Szpiro-shaped close

The claim’s architecture — a topological quantity evaluated once as a sum of local contributions and once as a globally protected obstruction, with an inequality falling out of the comparison — is the architecture of the geometric Szpiro inequality [REF: Amorós-Bogomolov-Katzarkov-Pantev; Zhang; Joshi §10 for the presentation we used]. The dictionary, with its honest entries (CT-4, graduated 2026-06-11):

geometric Szpiro	this paper
relation $[[a_j, b_j] \cdot \prod y_s = 1 \text{ in } \pi_1$ monodromy in $SL_2(\mathbb{R})$	torus plaquette relations core holonomy in $SO(2)$, carried by partial isometries
central extension; height/translation quasimorphism	universal cover $\mathbb{R} \rightarrow SO(2)$; lifted angle, an honest homomorphism
central element evaluated two ways discriminant $\leq C \cdot (2g - 2 + S)$	winding: Σ local fluxes vs flatness's zero $2\pi k/m^2$ vs δ^* and $c(k)$: the crossover inequality

The one honest discrepancy is also where the analogy bit back: Zhang's load-bearing input is quasimorphism defect control (Milnor-Wood), genuinely hyperbolic; our core is abelian and the lift is a homomorphism — but the first candidate replacement (gap-protected lift continuity over the column-path winding) was **falsified, machine-checked** (the shear family: a winding-changing, gap-preserving deformation at vanishing cost — exactly the kill criterion this thread had pre-registered). What must replace defect control is index stability for the magnetic-translation pair, where the defect is the sup of plaquette holonomies — a pair now machine-checked (`MagPair.lean`), with $\|[U, V]\| \leq 2|\sin(\pi k/m^2)|$. The shape transfers; the hard part — the index's rank-uniform stability for partial isometries — is new and now maximally sharp: the dilation gate (2026-06-13) isolated it as the sole non-classical content, and the machine-checked reduction (2026-07-01) compressed the entire remaining distance between evidence and theorem into the one named estimate `FluxPairRigidity`, with everything on either side of it — the assembly of flat families into commuting contraction pairs, the m -free locality constant, the classical Voiculescu core at full rank — proved. That is the correct division of labor between an analogy and a theorem, and it is where this paper stops and the mathematics continues.

8. Open problems

- (1) Stage B — the m -uniform rigidity lemma (the program's central open problem).
- (2) A matrix-valued two-regime law (F2): does δ^* compose the same way when the carried quantity is an operator block, and does the regenerative bonus survive?
- (3) The $c(k)$ -spectrum: is $c(k)$ monotone in k ? bounded? ($m^* \propto \sqrt{k}$ says detectability worsens with charge; does protection strengthen? Partial answer, three-gaps §4a + `expM`: at $\text{codim} \geq 2$ the constant is exactly 1, k -independent — protection neither strengthens nor weakens with charge there; the $\text{codim}-1$ $\sqrt{2}$ is likewise measured k -uniform. So detectability degrades with k while protection stays flat: the stealth window *widens* with charge.)
- (4) A converse stealth statement: can *every* sub-threshold unremovable deformation be realized by a topological charge, or are there non-topological stealth classes?
- (5) The $r = 1$ boundary as the bridge to the rank-one companion world. (Updated 2026-07-03 — the cell was run and the expectation “no stealth at $r = 1$ ” was wrong: the only invariant there is $\mathbb{Z}/2$ frustration, and it suffices — a curvature-free wound family is locally invisible yet plateaus at height exactly 1 — **PROVED 2026-07-04**, both bounds, closed-form per-edge alignment costs (`./paper2-r1-theorem-draft.md`; falsification map, row 3). The rank-one sector is thereby the first with an end-to-end-theorem stealth statement: local defect zero, sub-threshold for every instrument, unremovable below cost exactly 1 = 35×

the worked δ^* . Remaining there: the noisy ($\sigma > 0$) window, and the Lean formalization.)

Figure plan: F1 the stealth-regime demonstration (planned; §6); F2 the crossover diagram — $\delta\text{-local}(m) = 2\pi k/m^2$ falling through δ^* with $c(k)$ flat above (renderable now from e2v2 npz + E3 constants); F3 the Szpiro dictionary as a two-column graphic; T1 the falsification map. Word count ≈ 2400 . Venue thought: arXiv math.OA cross-list physics.data-an, or Experimental Mathematics as a companion-pair with Paper 2. (Status 2026-07-01: the IP freeze was lifted 2026-06-13; outward moves gated only by ordinary confirm-before-irreversible care and the third-party name scrub.)

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[1].) - **Zhang:** S.-W. Zhang, *Geometry of algebraic points*, in: First International Congress of Chinese Mathematicians (Beijing, 1998), AMS/IP Stud. Adv. Math. **20**, AMS/International Press (2001), 185–198. (§3: the geometric Szpiro inequality via monodromy in SL_2 + the translation quasimorphism on $\tilde{SL}_2(\mathbb{R})$ — the proof §7’s dictionary transcribes, verified line-for-line.) - **Joshi §10:** K. Joshi, *Construction of Arithmetic Teichmüller Spaces II½: Deformations of Number Fields*, arXiv:2305.10398, §10 (“Appendix: The proof of geometric Szpiro inequality revisited” — the presentation we used; reading note provenance: v12, §§10.1–10.18, display (10.15.1)). - Milnor-Wood (the quasimorphism/relative-Euler-number framing): J. Milnor (1958), J. W. Wood (1971).

Figures (print appendix)

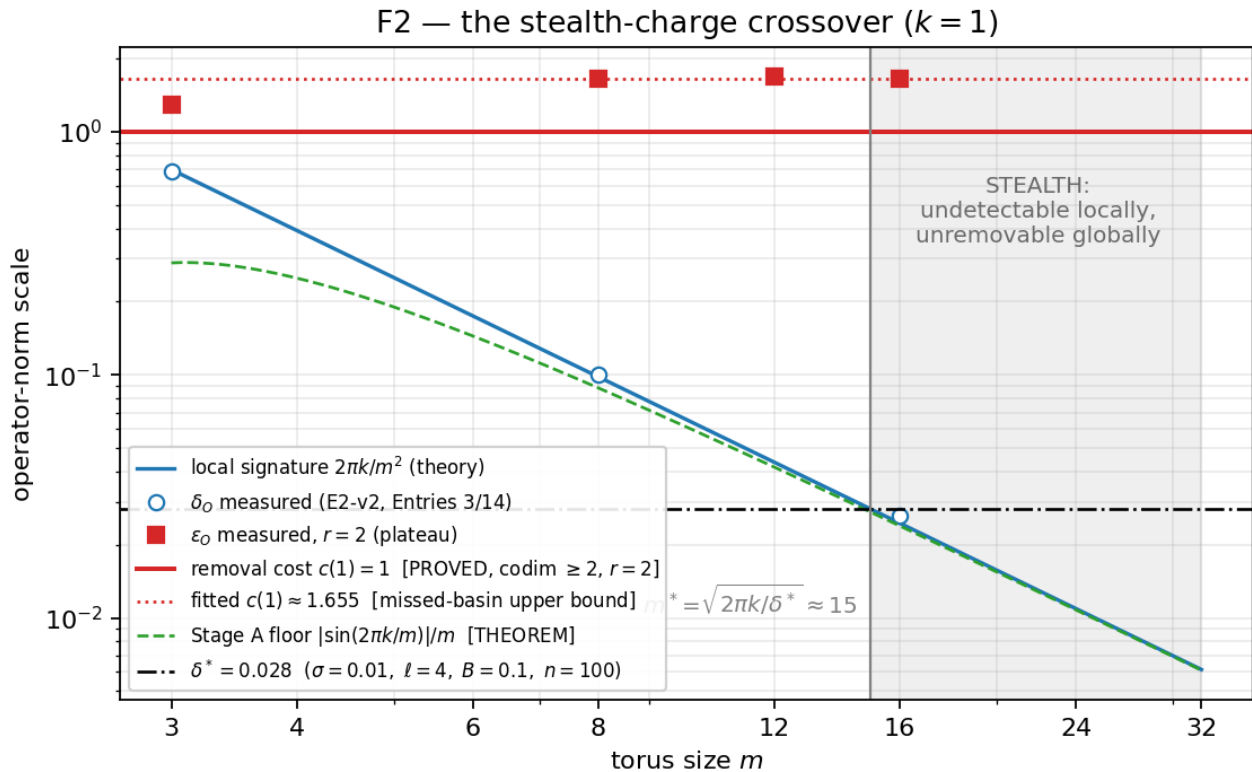


Figure 1: F2 — The crossover: local signature $2\pi k/m^2$ falls through δ^* at m^* ; the removal floor (proved, $\text{codim} \geq 2$: exactly 1) stays flat above; fitted 1.655 shown dashed as a missed-basin upper bound.

Authorship and AI-assistance disclosure

This research program — its conception, its original ideas, its questions, and its direction — is the work of the author, Juan Carlos Paredes. The strategic pivots, the falsification discipline, the decisions about what to pursue, what to test, what to claim, and what not to claim are his throughout. In the course of the program he directed large language

Stealth-regime demonstration (synthetic, planted charge $k=1$): a topological charge invisible to every local test

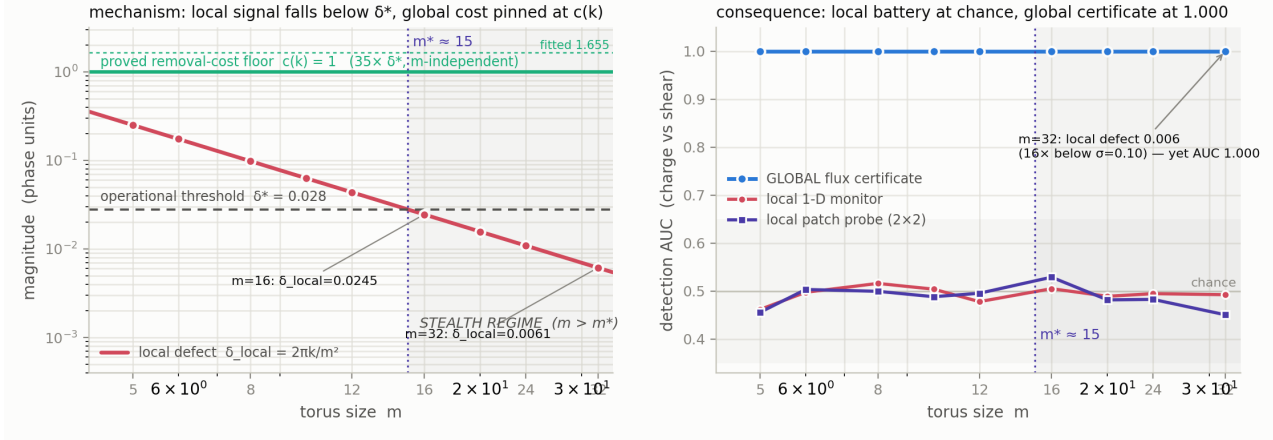


Figure 2: F1 — Stealth-regime demonstration (planted charge at $m > m^*$): local battery at chance, global fit cost pinned at the floor. Synthetic/planted experiment, per Sec. 6.

models (Anthropic Claude — Opus and Fable model families) as research assistants: implementing and running numerical experiments he commissioned, formalizing proofs in Lean 4, drafting and revising prose, and retrieving literature, all under his direction and review. Formal results are machine-checked (Lean 4) where stated; numerical claims are reproducible from the released scripts. Juan Carlos Paredes is the author of this work; the assistance is disclosed in the spirit of the full transparency this venue requires.